

CHERNYSHEVSKY, Russian Federation

WMO: 247240

Lat: 63.0171N

Lon: 112.4340E

Elev: 1383

StdP: 13.98

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-48.4	-43.4	-52.2	0.3	-48.4	-47.7	0.4	-43.4	21.1	-7.0	19.9	-12.0	7.9	230	0.742

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.5	82.0	64.0	78.2	62.0	74.6	60.3	65.7	78.2	63.6	74.8	61.7	72.3	7.1	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.2	84.8	70.6	59.1	78.6	68.2	57.0	72.8	66.6	31.3	78.5	29.7	75.1	28.3	72.5	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.3	17.1	15.4	DB	-48.5	88.1	6.1	2.4	-52.9	89.8	-56.5	91.2	-59.9	92.5	-64.4	94.3	(4)
(5)			WB	-48.5	68.9	6.0	2.3	-52.8	70.6	-56.3	71.9	-59.6	73.2	-63.9	74.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.8	-21.2	-15.3	4.9	23.9	39.7	58.6	63.7	56.7	41.1	20.8	-6.3	-20.0
	DBStd	32.43	13.92	13.08	13.35	10.31	9.49	9.59	7.90	7.26	8.41	10.80	14.99	14.51
	HDD50	11669	2206	1827	1397	782	344	32	4	24	286	906	1690	2171
	HDD65	16285	2671	2247	1862	1232	785	234	120	270	717	1371	2140	2636
	CDD50	995	0	0	0	0	24	290	429	232	20	0	0	0
	CDD65	135	0	0	0	0	0	42	80	13	0	0	0	0
	CDH74	884	0	0	0	0	1	280	527	76	0	0	0	0
	CDH80	188	0	0	0	0	0	55	121	12	0	0	0	0
Wind	WSAvg	7.8	8.8	7.9	8.0	8.3	8.1	6.9	6.2	6.8	7.5	8.4	8.2	8.5
Precipitation	PrecAvg	15.4	0.6	0.5	0.7	0.9	1.3	2.2	1.7	2.1	1.9	1.9	1.1	0.7
	PrecMax	25.0	1.3	1.0	1.0	1.5	2.5	4.5	4.3	5.8	4.0	3.5	1.8	1.2
	PrecMin	9.1	0.3	0.2	0.2	0.2	0.5	0.7	0.3	0.3	0.3	0.6	0.5	0.2
	PrecStd	3.2	0.2	0.2	0.3	0.4	0.5	1.1	0.9	1.1	0.9	0.8	0.4	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.8	22.1	39.5	51.3	70.2	85.7	87.7	81.8	67.6	45.0	30.3	22.1
		MCWB	13.9	21.2	32.0	39.4	51.9	64.1	67.8	63.7	54.5	39.4	28.9	21.4
	2%	DB	7.8	14.4	31.7	46.9	64.2	80.6	83.5	75.6	61.1	40.9	27.2	12.8
		MCWB	6.8	13.4	27.1	36.7	49.3	62.0	65.4	60.8	50.3	37.2	26.0	12.0
	5%	DB	3.3	9.3	27.1	42.6	59.1	76.9	79.7	71.7	56.5	37.1	22.4	5.6
		MCWB	2.7	8.4	23.6	34.0	46.1	60.3	63.2	59.1	48.0	33.8	21.3	4.8
	10%	DB	-2.5	3.8	23.0	38.4	54.2	73.3	76.3	68.1	52.5	33.9	15.8	0.1
		MCWB	-3.1	3.0	20.3	31.5	43.0	58.4	61.8	57.1	46.1	31.6	15.0	-0.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.9	21.2	32.3	39.8	53.5	66.2	69.8	65.5	55.2	40.4	29.6	21.5
		MCDB	14.4	22.3	38.1	50.4	67.2	81.7	85.2	77.8	65.4	43.9	30.3	22.2
	2%	WB	7.1	13.5	27.6	37.2	50.0	63.7	67.0	62.7	52.2	37.6	26.0	12.1
		MCDB	7.8	14.4	31.8	46.0	62.8	77.3	79.9	72.0	58.6	40.9	27.1	12.9
	5%	WB	2.6	8.4	23.9	34.6	46.8	61.6	64.9	60.4	49.5	34.6	21.4	5.0
		MCDB	3.2	9.2	26.9	41.3	58.1	74.3	76.2	68.9	54.5	36.6	22.5	5.7
	10%	WB	-3.1	3.0	20.4	32.2	43.8	59.6	63.0	58.5	47.0	31.8	14.9	-0.6
		MCDB	-2.5	3.6	23.0	37.8	53.3	70.8	73.6	66.2	52.0	33.5	15.8	0.0
Mean Daily Temperature Range	5% DB	MDBR	9.7	10.4	13.5	13.7	13.6	15.6	14.5	13.1	10.3	8.1	9.5	9.8
		MCDBR	14.1	14.6	14.1	15.1	19.4	18.2	18.0	17.8	15.5	9.7	13.9	14.8
		MCWBR	13.9	14.4	11.5	9.9	11.4	8.5	8.1	8.3	8.5	7.8	13.9	14.7
	5% WB	MCDBR	14.1	14.6	14.0	14.2	18.9	17.3	16.7	15.9	13.6	8.9	14.2	14.5
		MCWBR	14.0	14.3	11.7	9.9	11.5	8.4	8.5	8.5	8.9	7.9	14.0	14.4
Clear-Sky Solar Irradiance	taub		0.210	0.256	0.289	0.320	0.348	0.388	0.430	0.419	0.314	0.271	0.204	0.186
	taud		1.918	2.056	2.084	2.135	2.328	2.342	2.207	2.181	2.492	2.298	1.942	1.872
	Ebn at Noon		128	214	253	272	275	264	246	236	252	213	128	70
	Edh at Noon		11	22	32	38	35	35	39	37	22	17	10	5
All-Sky Solar Radiation	RadAvg		88	379	928	1561	1728	1816	1653	1236	741	332	119	38
	RadStd		5	22	38	81	107	137	109	87	74	29	11	2

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+2.25	N/A	N/A	+3.60	+0.85	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	+2.25	N/A	N/A	+3.60	+0.85	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page